

Statement of Work
For
Thermal Vacuum Chamber

Title: Thermal Vacuum Chamber and New Chiller Servicing and Installation

Document Date: 21 February 2025.

1. Scope:

The work performed will provide servicing to a thermal vacuum chamber to ensure it is capable of meeting the NASA GEVS and MIL STD 1540D/MIL HDBK 340A standards for flight qualifying space hardware, maintaining <0.1 mTorr and 10 degrees margin on the allowable flight temperature range, typically within -49 to 66 degrees Celsius. This work additionally requires the servicing of an upgraded chiller capable of supporting the aforementioned temperature range in the thermal vacuum chamber.

The contractor must find and resolve leaks preventing the chamber from achieving vacuum levels below 0.1mTorr.

The control panel for the system must be upgraded to incorporate a new processing unit and screen for the controller unit, compatible with supportable operating systems, such as Windows 11. A new temperature controller, compatible with the SnowRiver C80-6 chiller must also be incorporated.

The existing Chiller unit must be recharged with refrigerant, and checked to ensure it is not leaking either refrigerant or coolant, then integrated with the new control system to verify operation to a temperature range of at least -49C to +66C.

Custom crates for the chamber/solar simulator and chiller units shall be provided so that the units can be shipped to the contractor location for work without damage.

1.1 Background:

The documents below describe the requirements for TVAC testing to be implemented using this hardware. These documents drive the vacuum level and temperature range requirements for the system (<0.1mTorr, -49C to +66C temperature range).

2. Applicable Documents:

GSFC-STD-7000B - General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects

MIL STD 1540D (Department of the Air Force Standard Practice: Product Verification Requirements for Launch, Upper Stage, and Space Vehicles)

MIL HDBK 340A (Department of Defense Handbook: Test Requirements for Launch, Upper Stage and Space Vehicles, Vol 1: Baselines. Specifically chapter 7 and 8, Table XI {page 82} and paragraph 8.3.3.c {page 103})

Aerospace Report No. TR-2004(8563)-1 Rev A (Test Requirements for Launch, Upper Stage and Space Vehicles; Paragraph 6.3.9.3, page 56)

3 Requirements/Salient Characteristics:

3.1 Minimum Requirements: Thermal Vacuum: AFIT ABESS TVAC System, Serial number 51907

1. The chamber must be restored to a capability of maintaining a pressure of <0.1 mTorr. Reworking seals and fittings to ensure no air leaks into the chamber.
2. The chamber/chiller must be restored to a capability of maintaining a minimum temperature of -49 degrees Celsius.
3. The chamber/chiller must be restored to a capability of maintaining a maximum temperature of $+66$ degrees Celsius.
4. The solar simulator attached to the chamber must be restored to full design capability, providing 1 Sun of illumination (about 1353 Watts/m² of irradiated power) to a minimum circular area of 0.126 m².
5. The contractor shall refurbish the control system for the TVAC chamber to a modern/supportable operating system (Windows 11 meets those requirements).
6. The coolant lines shall be checked to ensure they do not leak and have sufficient capacity to properly operate the chamber to the temperature limits specified above.

3.2 Minimum Requirements: Chiller: ABBESS, SN 51211

1. The chiller shall be restored to full design operating capacity.
2. The contractor shall replenish the chiller refrigerant.
3. The contractor shall ensure no refrigerant or coolant leaks in the chiller before returning the chiller to AFIT.

3.3 Minimum Requirements: Shipping/Crating:

1. The contractor shall provide fitted crates to allow shipment of the chamber/solar simulator and the chiller to the contractor's facilities without damage.

3.4 Period of Performance:

Crating and shipment back to the contractor facility shall be completed within 60 days of contract award

Repairs, upgrades, integration and testing shall be complete within 10 weeks after arrival at the contractor facility.

Return to AFIT shall be completed within 30 days of repair/upgrade/integration/testing work completion.

3.5 Payment:

Payment should be made using Wide Area Work Flow, WAAF.